



Kanakku - Project Overview

Apple-Premium Personal Finance & Budget Management Platform

EXECUTIVE SUMMARY

Kanakku is a next-generation personal expense tracker and budget management application designed with Apple-grade aesthetics. Built with modern web and mobile technologies, it delivers real-time cloud synchronisation, AI-powered financial advisory via Google Gemini, Need vs. Want behavioural expense categorisation, and enterprise-grade PIN lock security.

1. Technical Stack & Infrastructure

Core frameworks, libraries, and runtime architecture

Category	Technology / Library	Version	Purpose & Key Benefits
Frontend Core	React + TypeScript + Vite	19.0 / 5.8 / 6.2	High-performance UI rendering, strict type safety, fast HMR build setup.
Styling & Motion	Tailwind CSS v4 + Framer Motion	4.1 / 12.2	Utility-first modern styling system with fluid 60fps animations.
Mobile Wrapper	Capacitor (@capacitor/android)	8.5	Cross-platform native Android container and bridge for APK builds.
Backend & DB	Firebase Auth + Cloud Firestore	12.18	Real-time multi-device sync, UID-isolated data storage, offline cache.
Artificial Intelligence	Google GenAI SDK (@google/genai)	2.4	Gemini AI budget coaching, financial analysis, voice input parsing.
Export & Reports	jsPDF, jsPDF-AutoTable, XLSX	4.2 / 5.0 / 0.18	Professional PDF statement generator & Excel spreadsheet exports.
Icons & UI Components	Lucide React, React-Image-Crop	0.54 / 11.1	Crisp SVG iconography & profile/receipt image cropping.

2. Key Features & Business Capabilities

- **Triple Transaction Model**
Full support for Expenses, Incomes, and Multi-account Transfers (Bank to Cash, Wallet, Savings).
- **Need vs. Want Behavioural Analytics**
Mandatory tagging of discretionary spending allowing users to identify lifestyle inflation and savings leaks.
- **Gemini AI Financial Coach**
Real-time AI consultation analyzing historical cashflow and suggesting personalized budget optimizations.
- **Voice & Smart Parser**
Natural language processing for converting raw voice or text (e.g., "Spent 450 on fuel from HDFC") into structured entries.
- **Security & App Lock**
Custom 4-digit PIN lock screen, session timeout management, and client-side biometric toggles.
- **Cross-Format Statements**
Automated generation of audit-ready PDF financial reports and Excel workbooks.

Kanakku — Architecture & Directory Layout

SYSTEM STRUCTURE

3. Repository & Source Code Structure

Path	Module / Component	Description
src/App.tsx	Application Shell	Main container handling lazy routing, security lock screen, and onboarding gate.
src/context/AppContext.tsx	Global State Manager	Central React context managing real-time Firestore listeners, auth, & transactions.
src/services/firestoreService.ts	Firestore Database API	UID-isolated CRUD methods, listeners for expenses, income, transfers, and profiles.
src/types/index.ts	Type Definitions	Strong TypeScript interfaces for Transactions, Locations, Budgets, and Settings.
src/components/views/DashboardView.tsx	Dashboard View	Financial summary cards, balance breakdown, trend charts, and recent activity.
src/components/views/InsightsView.tsx	Insights & Analytics	Category spending graphs, Need vs. Want distribution, AI Coach launcher.
src/components/views/BudgetView.tsx	Budget & Targets	Monthly spending limits per category, progress meters, and threshold alerts.
src/components/views/ProfileView.tsx	Profile & Accounts	User details, linked bank accounts, wallet balances, and financial health score.
src/components/views/SettingsView.tsx	Settings & Preferences	Currency selector, Dark mode toggle, PIN management, JSON Backup/Restore.
src/components/views/OnboardingView.tsx	Onboarding Wizard	Step-by-step introduction for initial balance, custom categories, and setup.
src/components/views/AddTransactionModal.tsx	Transaction Creator	Modal for rapid manual or voice-assisted logging of expenses and incomes.
src/components/modals/PinLockScreen.tsx	Security Lock Screen	Interactive numeric keypad for PIN authorization and app protection.
src/utils/voiceParser.ts	NLP Voice Parser	Regex and heuristic engine for parsing spoken phrases into transaction data.
src/utils/exportUtils.ts	Export Engine	Generates customized PDF statements and structured Excel workbooks.
android/	Android Native Project	Capacitor Android wrapper configured for building native mobile APKs.

4. Data Model & Schema Overview

Key TypeScript entities used across the application

Interface / Entity	Key Fields	Notes & Relationships
Transaction	id, type, amount, date, timestamp, userId, needWant, locationId, category, source	Polymorphic type: Expense, Income, or Account Transfer.

Interface / Entity	Key Fields	Notes & Relationships
MoneyLocation	id, name, type (cash/bank/wallet), initialBalance, isSavings, mask, institution	Represents a physical or digital account where balance is tracked.
ExpenseCategory	id, name, icon, bgColor, textColor, color, defaultNeed	Custom spending category with default Need vs. Want flag.
CategoryBudget	category, limit, userId	Monthly threshold for monitoring category spending.
UserProfile	name, age, profession, email, phone, memberSince, avatarUrl	User demographic and account profile.
AppSettings	currency, darkMode, isPinLockEnabled, pinCode, pushAlerts	Device and user configuration preferences.

5. Security, Storage & Offline Architecture

Kanakku prioritizes privacy and security with multi-tiered protection mechanisms:

- User Data Isolation (UID Sandboxing)**
Every Firestore query and mutation is strictly filtered by the authenticated user ID (UID), ensuring zero cross-tenant data access.
- Offline-First Capabilities**
Firestore offline persistence enables users to log transactions, view balances, and review budgets without an active internet connection; sync occurs seamlessly upon reconnection.
- Client-Side PIN Lockdown**
Enabling PIN lock safeguards financial records from unauthorized local access. Unlocks require entering the encrypted 4-digit passcode.
- Sanitized State & Exports**
All exports (PDF / Excel / JSON backups) are sanitized on-device without passing through unverified intermediary servers.

6. Build, Development & Deployment Commands

Target / Action	Command	Description
Start Dev Server	npm run dev	Launches Vite development server at http://localhost:3000 with HMR.
Production Build	npm run build	Builds optimized static assets into the dist/ directory.
TypeScript Check	npm run lint	Runs tsc --noEmit to ensure 100% strict type safety.
Preview Build	npm run preview	Locally serves the production bundle for pre-release testing.
Sync Android	npx cap sync android	Copies web assets and syncs Capacitor plugins with Android project.
Open in Android Studio	npx cap open android	Launches native Android project for APK / App Bundle generation.

DOCUMENTATION SUMMARY

Generated automatically for Kanakku Financial Application. All architectural guidelines and schemas are synchronized with the active codebase.